

1. Administrative & Study Identification

Full Study Title: A Phase 1/2, Multicenter, Single-arm, Open-label Study to Evaluate the Efficacy and Safety of Fedratinib in Japanese Subjects With DIPSS (Dynamic International Prognostic Scoring System)-Intermediate or High-risk Primary Myelofibrosis (PMF), Post-polycythemia Vera Myelofibrosis (Post-PV MF), or Post-essential Thrombocythemia Myelofibrosis (Post-ET MF) **Short Title/Acronym:** Fedratinib in Japanese Subjects with MF **Protocol Number:** [Internal Study ID] **NCT Number:** NCT04446650 **Trial Status:** Active, not recruiting **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** Fedratinib (Inrebic) **ATC Code:** L01EJ02 **Phase of Development:** Phase 1 / Phase 2 **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of Fedratinib to determine the Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D) in Phase 1, and to evaluate the overall efficacy (Response Rate) and safety of Fedratinib in Phase 2. **Secondary Objectives:** To evaluate Adverse Events (AEs) and assess the pharmacokinetics (Cmax, AUC) of Fedratinib in the study population.

2.2 Background & Rationale To address the therapeutic needs of Japanese subjects with DIPSS intermediate or high-risk primary myelofibrosis (PMF), post-PV MF, or post-ET MF. This study provides a targeted clinical evaluation for patients requiring intervention, including those who have previously received or discontinued ruxolitinib due to lack of efficacy or intolerability, to ascertain optimal dosing (300 mg or 400 mg) and safety profile in this demographic.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Dose escalation utilizing a mTPI-2 design in Phase 1)

3.2 Planned Enrollment & Duration Target \$N\$: 31 evaluable patients **Number of Sites:** Multicenter (Japan) **Estimated Study Start:** [Not Provided] **Estimated Primary Completion:** [Not Provided]

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 20 years with an ECOG Performance Status of 0, 1, or 2. 2. Diagnosis of PMF, post-PV MF, or post-ET MF with a DIPSS Risk score of Intermediate-1 (with symptoms), Intermediate-2, or High. 3. Measurable splenomegaly demonstrated by a spleen volume of ≥ 450 cm³ by MRI/CT, or palpable spleen measuring ≥ 5 cm below the left costal margin. 4. *Phase 1 specific*: Must have previously received ruxolitinib treatment for at least 14 days (or < 14 days if discontinued due to intolerability). (Note: Phase 2 allows patients who have never received ruxolitinib if they are expected to derive clinical benefit).

4.2 Exclusion Criteria 1. Laboratory abnormalities including: Platelets $< 50 \times 10^9$ /L, ANC $< 1.0 \times 10^9$ /L, WBC $> 100 \times 10^9$ /L, Myeloblasts $\geq 5\%$, CrCL < 30 mL/min, Serum amylase or lipase $> 1.5 \times$ upper limit of normal (ULN), AST or ALT $> 3 \times$ ULN, and Total bilirubin $> 1.5 \times$ ULN (unless direct bilirubin is $< 25\%$ of total). 2. Previous splenectomy or previous/planned hematopoietic stem-cell transplantation (SCT). 3. History or presence of encephalopathy (including Wernicke encephalopathy) or signs/symptoms of encephalopathy without documented exclusion. 4. Uncorrected thiamine deficiency in whole blood.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Fedratinib continuous daily dosing (exploring 300 mg and 400 mg doses in Phase 1 to establish the RP2D). **Route of Administration**: Oral capsule **Duration of Treatment**: Continuous through 3 periods: Screening, Treatment (until progression/unacceptable toxicity) including a 30-day follow-up, and Survival Follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for myelofibrosis and comorbidities. **Prohibited**: Strong/moderate CYP3A4 inducers, dual CYP2C19/CYP3A4 inhibitors, concurrent chemotherapy, immunomodulatory drugs (e.g., thalidomide, interferon-alpha), anagrelide, systemic corticosteroids > 10 mg/day prednisone, and ruxolitinib (within 14 days of start).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day -1	Informed Consent, Thiamine tests, Baseline MRI/CT for Spleen Volume, Lab panels
Baseline	Day 1	Eligibility confirmation, First Dose of Fedratinib
Interim	End of Cycle 6	Efficacy Assessment (Spleen Volume Reduction), Safety Labs, PK Assessments
End of		

Study | Post-Treatment | 30-day safety follow-up after last dose;
Survival follow-up period |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 1. *Phase 1:* Maximum Tolerated Dose (MTD) based on DLTs in the first cycle; Recommended Phase 2 Dose (RP2D). 2. *Phase 2:* Response Rate (RR), defined as the proportion of subjects who have $\geq 35\%$ Spleen Volume Reduction (SVR) at the end of Cycle 6 from baseline. **Measurement Method:** Central/Local Imaging (MRI or CT scan) and Safety Review Committee assessment.

7.2 Secondary & Exploratory Endpoints 1. Incidence and severity of Adverse Events (AEs). 2. Pharmacokinetics: Maximum observed concentration (C_{max}) of fedratinib. 3. Pharmacokinetics: Area under the concentration-time curve (AUC) of fedratinib.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous reporting with strict monitoring for neurological signs (e.g., ataxia, ocular paralysis) suggestive of encephalopathy. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline; specific tracking of thiamine levels and MRI evaluations for suspected Wernicke encephalopathy. **Data Monitoring Committee (DMC):** An internal Safety Review Committee will review Phase 1 dose-limiting toxicities (DLTs) to determine the MTD and RP2D.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) / Efficacy Evaluable Set, and Safety Set. **Sample Size Justification:** Total $N=31$. The Phase 1 portion leverages a modified Toxicity Probability Interval-2 (mTPI-2) design for robust and ethical dose escalation/de-escalation. **Handling of Missing Data:** Standard handling per institutional biostatistics guidelines for oncology trials (e.g., non-responders for early dropouts prior to Cycle 6 MRI).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with the International Council for Harmonisation (ICH) guidelines, Good Clinical Practice (GCP), and the Declaration of Helsinki. **Monitoring Plan:** Regular clinical monitoring, source document verification, and strict monthly documented verification of true abstinence or highly effective contraception for patients. **Audit**

Trail: eCRF locking and compliance tracking for laboratory processing (especially whole blood thiamine levels prior to treatment).

11. Study Identifiers & Governance

Primary ID (Posting ID): 130976435661858653 **Registry Number:** NCT04446650 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study of Fedratinib in Japanese Subjects With DIPSS (Dynamic International Prognostic Scoring System)- Intermediate or High-risk Primary Myelofibrosis (PMF), Post-polycythemia Vera Myelofibrosis (Post-PV MF), or Post-essential Thrombocythemia Myelofibrosis (Post-ET MF) **Official Regulatory Title:** A Phase 1/2, Multicenter, Single-arm, Open-label Study to Evaluate the Efficacy and Safety of Fedratinib in Japanese Subjects With DIPSS (Dynamic International Prognostic Scoring System)-Intermediate or High-risk Primary Myelofibrosis (PMF), Post-polycythemia Vera Myelofibrosis (Post-PV MF), or Post-essential Thrombocythemia Myelofibrosis (Post-ET MF) **EMA Information URL:** https://www.ema.europa.eu/en/documents/product-information/inrebic-epar-product-information_en.pdf

12. Clinical Framework (Myelofibrosis Specific)

Primary Condition: Primary Myelofibrosis (PMF), Post-PV MF, Post-ET MF **Condition Definition:** A chronic myeloproliferative neoplasm characterized by bone marrow fibrosis, proliferation of atypical megakaryocytes and granulocytes in the bone marrow, anemia, splenomegaly, and extramedullary hematopoiesis. (WHO, 2001) **Terminology Codes:** MeSH Code: D055728 | SNOMED ID: 128843003 **Diagnosis Threshold:** DIPSS Intermediate-1 (with symptoms), Intermediate-2, or High **Medical Methodology:** JAK2 Kinase Inhibition Therapy **Optimization Target:** $\geq 35\%$ Spleen Volume Reduction (SVR)

13. Study Procedures & Methodology

Management Pathway: Fedratinib clinical dosing pathway (mTPI-2 dose escalation) **Education Protocol (HCP):** Strict protocol training on continuous thiamine monitoring & encephalopathy risk assessment **Education Protocol (Patient):** Mandatory counseling on contraceptive requirements and monthly adherence tracking **Quality Audit Mechanism:** Centralized imaging reviews (MRI/CT) and stringent baseline laboratory screening

14. Observation & Follow-Up Schedule

Total Duration: Treatment Period + 30-Day Follow-up + Long-term Survival Follow-up **Onsite Visit Cadence:** Frequent monitoring during Cycle 1 for DLTs; periodic assessments up to Week 24 (End of Cycle 6) and beyond **Remote Monitoring Frequency:** Standard clinical trial remote auditing cadence **Checklist Review Interval:** Monthly review of abstinence/contraceptive compliance

15. Sign-off & Approval

Role	Name	Signature	Date						
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]						
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]						
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]						